Exp No: 29

NAME: J.Goutham Kumar Reddy

ROLL NO: 192311140

PRIORITY-BASED INTERRUPT HANDLING

AIM:

To model priority-based interrupt servicing.

SOFTWARE REQUIRED:

• Ubuntu (or WSL Ubuntu)

• g++

• CMake

• SystemC

PROGRAM:

```
#include <systemc.h>
SC_MODULE(priority_interrupt) {
sc_event low_int, high_int; void
low_ISR() { while(true) {
wait(low_int); cout << "Low Priority
Interrupt at " << sc_time_stamp() <<
endl; wait(3, SC_SEC);
}
}
void high_ISR() { while(true) {
wait(high_int); cout << "High
Priority Interrupt at "
```

```
<< sc_time_stamp() << endl; wait(1,
SC_SEC);
}
}

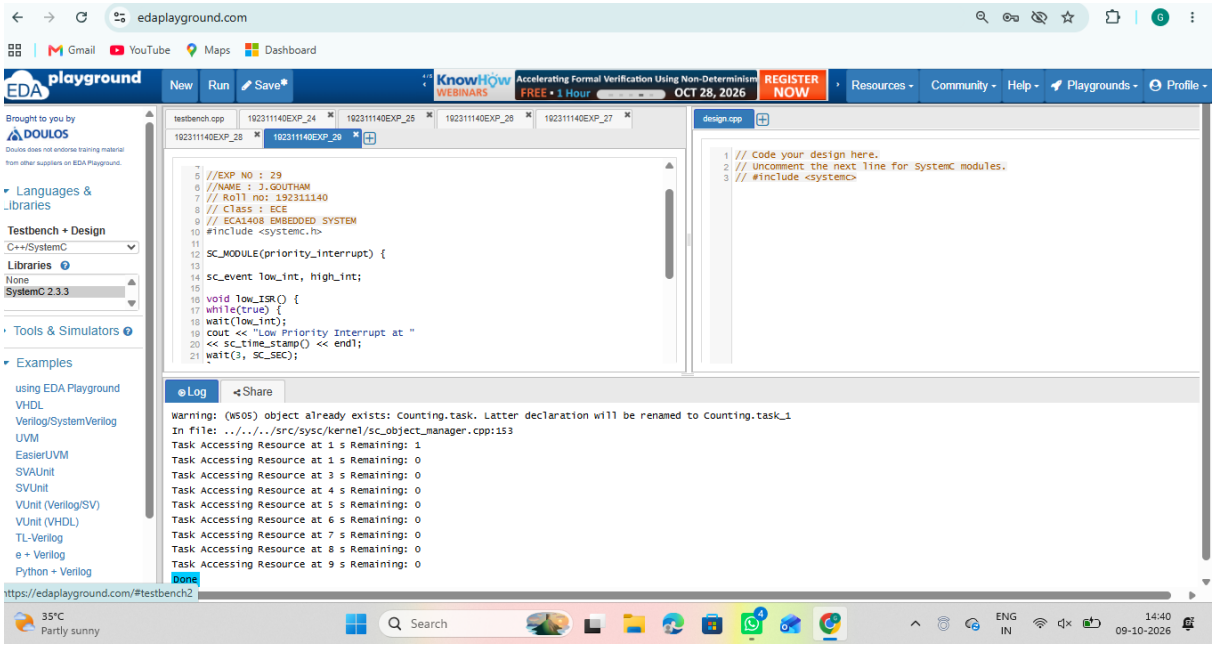
void generator() { wait(1,
SC_SEC);
low_int.notify();

wait(1, SC_SEC);
high_int.notify();
}

SC_CTOR(priority_interrupt) {
SC_THREAD(low_ISR);
SC_THREAD(high_ISR);
SC_THREAD(generator);
}
};

int sc_main(int argc, char* argv[]) {
priority_interrupt obj("Priority_INT");
sc_start(10, SC_SEC); return 0;
}
```

OUTPUT:



RESULT:

Thus the Priority based Interrupt handling was executed using SystemC and verified.